

JOAQUIN LOPEZ N.

[EMAIL](#) | [WEBSITE](#) | [GITHUB](#) | [LINKEDIN](#)

SANTIAGO, CHILE

PROFESSIONAL SUMMARY

Computer Engineer and software developer with experience building internal developer tools, backend systems, CI/CD automations, and network software. Proficient in Python, C, Rust, and SQL, with hands-on experience in Jenkins, GitHub Actions, databases, and Linux-oriented development workflows. Interested in backend, systems, and developer productivity engineering.

EDUCATION

University of Chile 2019 - 2025
B.S. | *Engineering Sciences, Major in Computing - 2024* Santiago, Chile
Professional Degree | *Computer Engineering - 2025*

TECHNICAL SKILLS

Languages: Python | C | Rust | C# | Java | JavaScript | Perl | Bash | SQL
Backend and data: Flask | Django | Pandas | NumPy | NetworkX | Hadoop | Spark
DevOps and tooling: Jenkins | GitHub Actions | Git | Perforce | P4 Code Review
Databases: PostgreSQL | MySQL | SQLite | Neo4j | MongoDB
Systems and embedded: FreeRTOS | ESP-IDF | Arduino | I2C

EXPERIENCE

Senior R&D Engineer | *Synopsys* May 2026 - Present
Infrastructure modernization and internal service ownership, DevTools team Santiago

- Drive infrastructure update projects and maintain widely used internal services.
- Design and implement CI/CD automations with Jenkins and GitHub Actions.
- Improve internal development workflows through tooling and process automation.

R&D Engineer | *Synopsys* Feb 2025 - Apr 2026
Internal tooling and engineering support, DevTools team Santiago

- Developed and maintained internal developer tools using Perl, Python, and Java.
- Diagnosed and resolved software and workflow issues reported by internal engineering teams.
- Maintained CI/CD workflows and provided technical support for engineering tools.

Software Developer | *Niclabs* Sep 2024 - Jan 2025
Development of an open-source DNS library in Rust Santiago

- Implemented RFC-defined functionality for a DNS stub resolver.
- Added DNSSEC authentication functionality and validated protocol behavior through testing and debugging.

Software Development Intern II | *CCHEN* Jan 2024 - Mar 2024
Development of a camera control system for plasma data acquisition Santiago

- Developed a camera control system for synchronized image capture during plasma experiments.
- Enabled real-time camera configuration and image transfer over Ethernet.

Software Development Intern I | *SPEL*
Embedded software development for CubeSats

Jan 2023 - Oct 2023
Santiago

- Ported a C-based CubeSat software platform to ESP32 and FreeRTOS.
- Migrated the communications layer to the LibCSP protocol library.
- Implemented a hardware abstraction layer for an I^2C gyroscope.

Teaching Assistant | *University of Chile, FCFM*
Teaching assistant for mathematics and computer science courses

2022 - 2024

- Courses taught: Systems Software Programming, Theory of Computation, Discrete Mathematics, Calculus, and Algebra.

PROJECTS

Thesis | *Semantic network extraction from dictionaries*

Jun

Extraction, analysis, and visualization of a semantic network from a Spanish synonym dictionary 2024 - 2025

- Built a Tesseract OCR pipeline to extract and process dictionary data from PDF documents.
- Generated a semantic network and stored its relationships in Neo4j.
- Developed a web application with vis.js to explore the network interactively.
- Analyzed network topology and structural properties with NetworkX.
- View the [project website](#).

Software Project: dnsrust | *Niclabs*

Mar 2024 - Aug 2024

Team development of an open-source DNS library in Rust

Santiago

- Collaborated in a five-person team to implement TSIG shared-key authentication.
- Implemented, tested, and debugged DNSSEC security extensions.
- View the [repository](#).

LANGUAGES

- English (C1 Advanced)
- Spanish (Native)